for example, evidence from Facebook in Altay et al. (2020)). Finally, social media users often engage in criticisms of available content and inform others about misinformation (see, for example, Kim et al. (2020) for evidence in the context of 2018 midterm elections).

3 Equilibria in General Networks

In this section, we characterize the structure of equilibria for any sharing network structure $\mathbf{P}$ and provide various comparative statics. Without loss of generality (and ease of exposition), we fix the article's message as $m = R$ for the remainder of the paper.[11]

3.1 Cutoff Strategies and Strategic Complementarities

When agent i receives an article with reliability r and message $m = R$, she updates her (*ex post)* belief, π_i, that the article is truthful according to Bayes' rule:

$$\pi_i = \frac{(pb_i + (1-p)(1-b_i))\phi(r)}{(qb_i + (1-q)(1-b_i))(1-\phi(r)) + (pb_i + (1-p)(1-b_i))\phi(r)}. \tag{2}$$

Clearly, π_i is increasing in b_i since an agent is more likely to believe in an article's veracity when its message agrees with her prior. Moreover, π_i is increasing in r, as the agent updates more on the basis of more reliable articles.

We can also see that the payoff to sharing ($\mathcal{S}$) increases in π_i, since the first component of utility, $U_i^{(1)}$, is increasing in π_i (as the individual would like to share truthful articles), while $U_i^{(2)}$ is independent of π_i. With a similar reasoning, the payoff to disliking ($\mathcal{D}$) is decreasing in π_i, whereas the payoff to ignoring ($\mathcal{I}$) is independent of π_i. This monotone behavior of payoffs will lead to best-response decision rules for agents that take the form of cutoff strategies, as we explain next.

We say that agent i employs a *cutoff strategy* if there exists $b_i^*(r)$ and $b_i^{**}(r)$ such that agent i chooses $\mathcal{S}$ when $b_i > b_i^{**}(r)$, chooses $\mathcal{I}$ when $b_i^*(r) < b_i < b_i^{**}(r)$, and chooses $\mathcal{D}$ when $b_i < b_i^*(r)$. Cutoff strategies in our context imply that agents who strongly agree with an article tend to share it, agents who strongly disagree with it tend to choose dislike, and those with intermediate beliefs typically ignore the article.

We will see in the next theorem that all equilibria will be in cutoff strategies. This means, in particular, that an equilibrium can be summarized by cutoff vectors $(\mathbf{b}^*, \mathbf{b}^{**}) = (b_1^*, b_1^{**}, \ldots, b_N^*, b_N^{**})$. Furthermore, these cutoffs $b_i^*(r)$ and $b_i^{**}(r)$ will both be decreasing in r, so that as reliability increases, an article becomes more likely to be shared and less likely to be disliked.

We can also note that our social media game exhibits *strategic complementarities*. To see this, observe that when others share more, meaning that b_i^{**} (weakly) decreases for all i, this tends to encourage more sharing for every agent j, because the second component of utility, $U_j^{(2)}$, increases, raising the overall utility of sharing. Similarly, when others reduce their likelihood of disliking, meaning that now b_i^* (weakly) decreases for all i, this reduces the likely cost of sharing misinformation by

[11] To see this is without loss of generality, observe that the analysis applies identically with an $m = L$ message but with complementary priors $b_i' = 1 - b_i$.